Cleaning and Tidying

Henry Heil, Jude Thomason

Loading Dataset

```
library(readr)
library(tidyverse)
library(modelr)
aptDataRaw <- read_delim("apartments_for_rent_classified_100K.csv",
    delim = ";", escape_double = FALSE, trim_ws = TRUE)

aptDataWorking <- aptDataRaw
```

Cleaning/Tidying

Some initial looks:

```
summary(aptDataWorking)
```

```
       id                  category           title               body          
 Min.   :5.121e+09   Length   :98924   Length   :98924   Length   :98924  
 1st Qu.:5.198e+09   N.unique :    7   N.unique :58272   N.unique :93978  
 Median :5.509e+09   N.blank  :    0   N.blank  :    0   N.blank  :    0  
 Mean   :5.358e+09   Min.nchar:   12   Min.nchar:   NA   Min.nchar:   NA  
 3rd Qu.:5.509e+09   Max.nchar:   30   Max.nchar:   NA   Max.nchar:   NA  
 Max.   :5.669e+09                                                        

    amenities         bathrooms        bedrooms          currency        
 Length   :98924   Length   :98924   Min.   :0.000   Length   :98924  
 N.unique : 9816   N.unique :   17   1st Qu.:1.000   N.unique :    1  
 N.blank  :    0   N.blank  :    0   Median :2.000   N.blank  :    0  
 Min.nchar:    2   Min.nchar:    1   Mean   :1.728   Min.nchar:    3  
 Max.nchar:  184   Max.nchar:    4   3rd Qu.:2.000   Max.nchar:    3  
                                     Max.   :9.000                    
                                     NAs    :123                      
       fee            has_photo        pets_allowed       price        
 Length   :98924   Length   :98924   Length   :98924   Min.   :  100  
 N.unique :    2   N.unique :    3   N.unique :    6   1st Qu.: 1013  
 N.blank  :    0   N.blank  :    0   N.blank  :    0   Median : 1350  
 Min.nchar:    2   Min.nchar:    2   Min.nchar:    4   Mean   : 1527  
 Max.nchar:    3   Max.nchar:    9   Max.nchar:   14   3rd Qu.: 1795  
                                                      Max.   :52500  
                                                      NAs    :1      
  price_display       price_type     square_feet          address         
 Length   :98924   Length   :98924   Min.   :  101.0   Length   :98924  
 N.unique : 3717   N.unique :    3   1st Qu.:  728.0   N.unique : 7772  
```

```
 N.blank  :    0   N.blank  :    0   Median :  900.0   N.blank  :    0
 Min.nchar:    4   Min.nchar:    6   Mean   :  956.4   Min.nchar:    4
 Max.nchar:   19   Max.nchar:   14   3rd Qu.: 1115.0   Max.nchar:   76
                                     Max.   :50000.0

     cityname           state          latitude       longitude
 Length   :98924   Length   :98924   Min.   :19.57   Min.   :-159.37
 N.unique : 2980   N.unique :   52   1st Qu.:33.75   1st Qu.:-104.79
 N.blank  :    0   N.blank  :    0   Median :37.23   Median : -84.57
 Min.nchar:    3   Min.nchar:    2   Mean   :36.95   Mean   : -91.57
 Max.nchar:   22   Max.nchar:    4   3rd Qu.:39.95   3rd Qu.: -77.61
                                     Max.   :64.83   Max.   : -68.78
                                     NAs    :25      NAs    :25
      source           time
 Length   :98924   Min.   :1.544e+09
 N.unique :   25   1st Qu.:1.551e+09
 N.blank  :    0   Median :1.569e+09
 Min.nchar:   NA   Mean   :1.560e+09
 Max.nchar:   NA   3rd Qu.:1.569e+09
                   Max.   :1.577e+09
```

```
unique(aptDataWorking$category)
```

```
[1] "housing/rent/apartment"         "housing/rent/home"
[3] "housing/rent/short_term"        "housing/rent"
[5] "housing/rent/condo"             "housing/rent/other"
[7] "housing/rent/commercial/retail"
```

```
unique(aptDataWorking$bathrooms)
```

```
 [1] "1"    "1.5"  "2"    "2.5"  "3"    "3.5"  "4"    "7"    "4.5"  "null"
[11] "5"    "8"    "8.5"  "6"    "5.5"  "9"    "7.5"
```

```
unique(aptDataWorking$bedrooms)
```

```
 [1]  1  3  2  4  0  5  7  8 NA  6  9
```

```
unique(aptDataWorking$currency)
```

```
[1] "USD"
```

```
unique(aptDataWorking$fee)
```

```
[1] "No"  "Yes"
```

```
unique(aptDataWorking$has_photo)
```

```
[1] "Thumbnail" "No"        "Yes"
```

```
unique(aptDataWorking$pets_allowed)
```

```
[1] "Cats"           "Cats,Dogs"      "None"           "Dogs"
[5] "null"           "Cats,Dogs,None"
```

```
unique(aptDataWorking$price_type)
```

```
[1] "Monthly"        "Weekly"         "Monthly|Weekly"
```

```
unique(aptDataWorking$state)
```

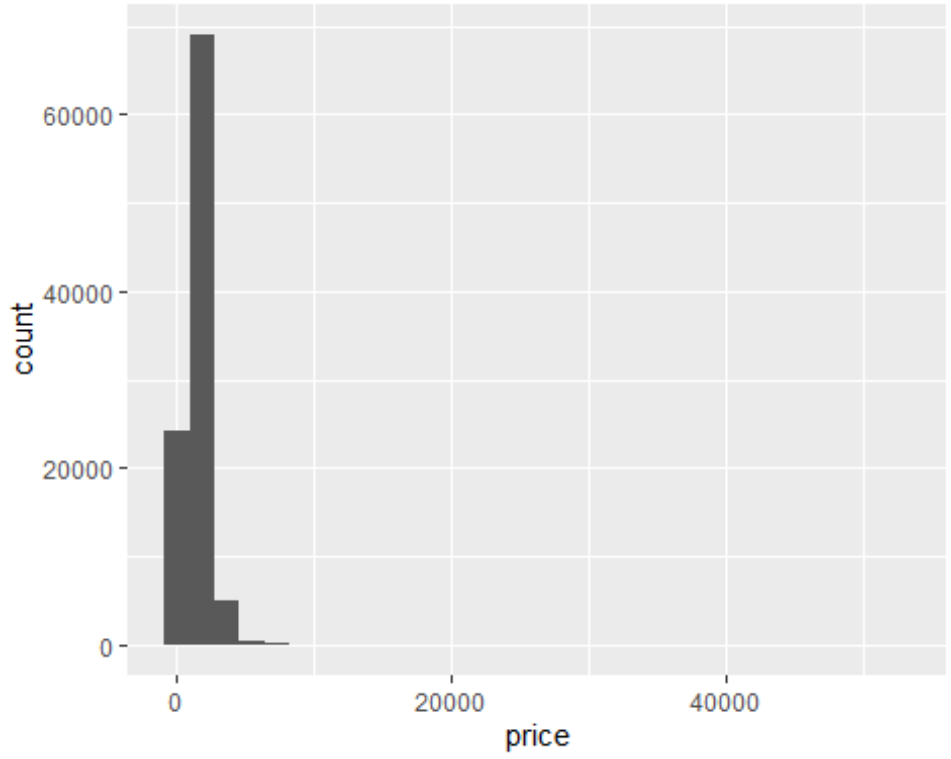
```
 [1] "CA"   "VA"   "NC"   "NM"   "CO"   "WV"   "GA"   "MA"   "DC"   "AZ"
[11] "IA"   "WA"   "TX"   "IL"   "MS"   "OR"   "FL"   "MO"   "PA"   "WI"
[21] "OK"   "UT"   "RI"   "NJ"   "IN"   "MD"   "OH"   "TN"   "ND"   "NE"
[31] "AR"   "MI"   "MN"   "HI"   "ID"   "SC"   "KS"   "AL"   "SD"   "NY"
[41] "KY"   "LA"   "AK"   "CT"   "NV"   "WY"   "VT"   "NH"   "MT"   "DE"
[51] "null" "ME"
```

```
unique(aptDataWorking$source)
```

```
 [1] "RentLingo"         "ListedBuy"         "RentDigs.com"
 [4] "RENTCaf\xe9"       "GoSection8"        "Listanza"
 [7] "RealRentals"       "RENTOCULAR"        "tenantcloud"
[10] "Real Estate Agent" "rentbits"          "Home Rentals"
[13] "Nest Seekers"      "RentFeeder"        "vFlyer"
[16] "Claz"              "Real Estate Shows" "Seattle Rentals"
[19] "BostonApartments"  "SpreadMyAd"        "Apartable"
[22] "Z57"               "FreeAdsTime"       "AgentWebsite"
[25] "HousesForRent"
```

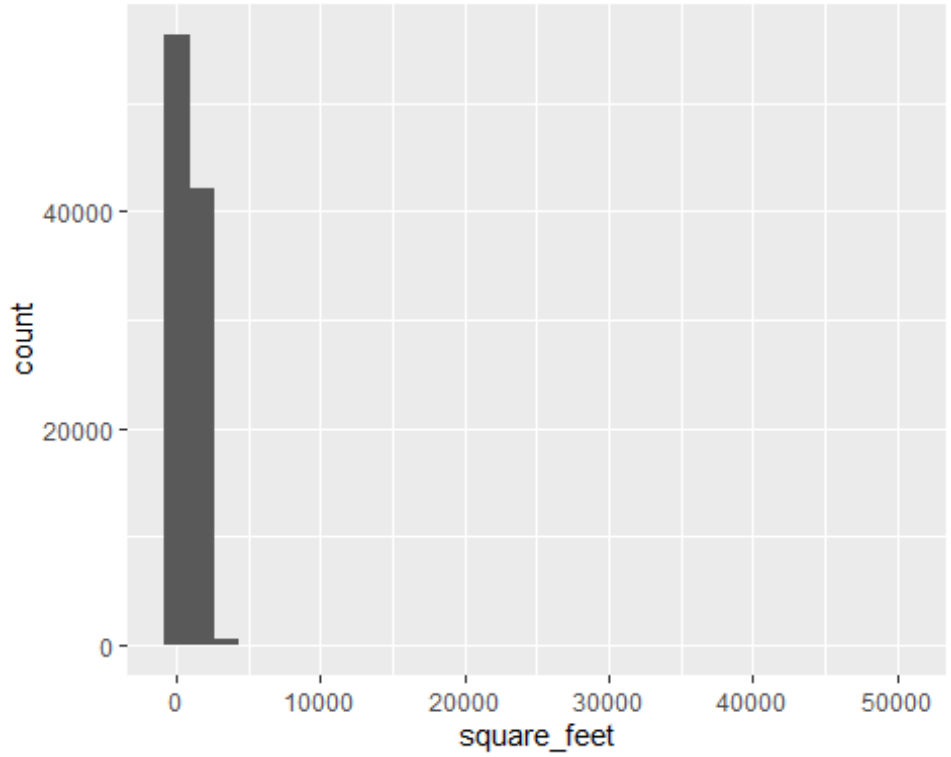
```
ggplot(aptDataWorking) +
  geom_histogram(mapping = aes(x=price))
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



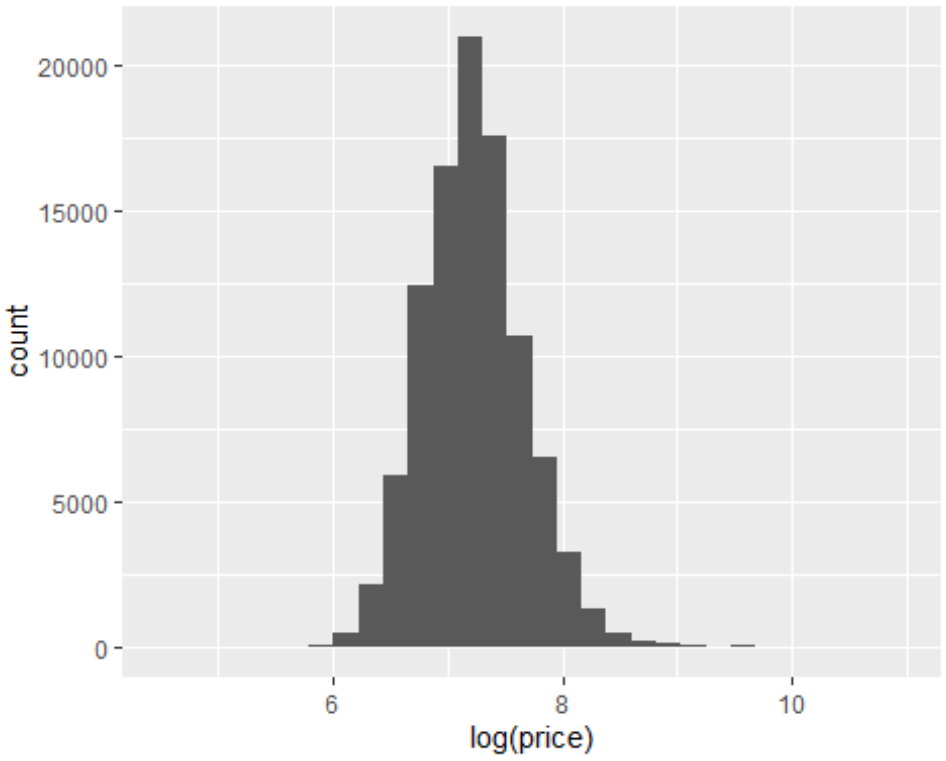

```
ggplot(aptDataWorking) +
  geom_histogram(mapping = aes(x=square_feet))
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



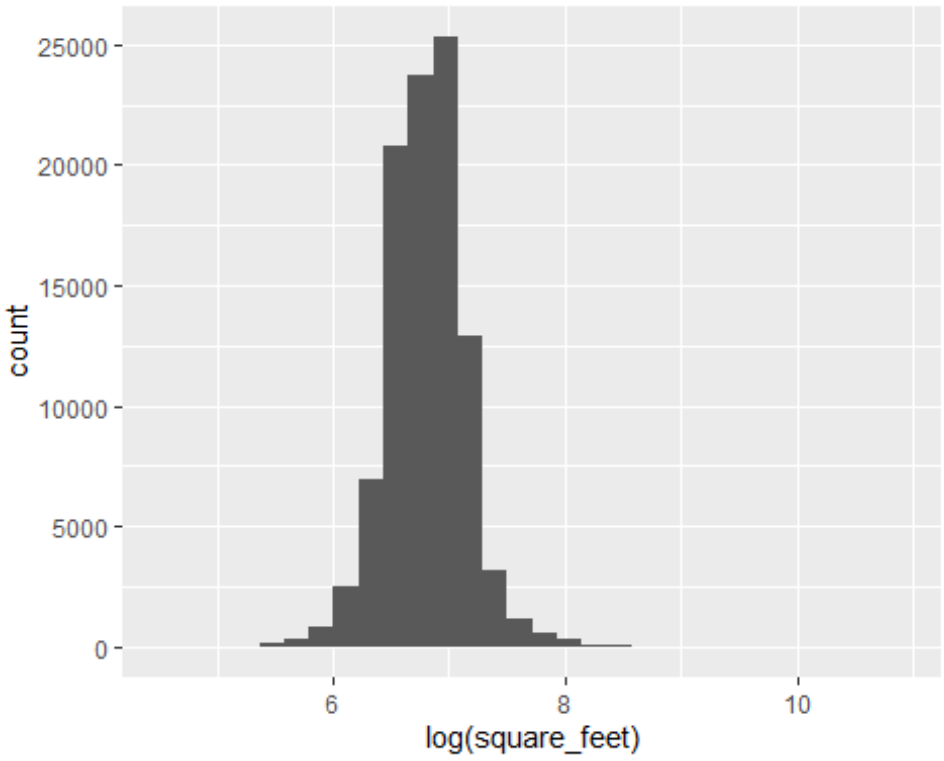

```
ggplot(aptDataWorking) +
  geom_histogram(mapping = aes(x=log(price)))
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(aptDataWorking) +
  geom_histogram(mapping = aes(x=log(square_feet)))
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Takeaways:

- amenities should be separated into columns, one for each amenity listed, and the same for pets_allowed.
- Most categorical variables should be made into factors, bedroom and bathroom should be made numerical
- Note that “DC” is one value in the state variable, which is reasonable
- There’s a glitch in the RentCafe values in source due to text formatting
- price and square_feet are extremely right-skewed, which can be corrected with logarithmic transformation
- Three price_type values are weekly instead of monthly. Converting their prices by multiplying by 52/12 seems reasonable. One price is in the enigmatic “Monthly|Weekly” format and seems discardable.

We start by fixing price_type.

Fixing weekly prices

```
aptDataWorking <- aptDataWorking %>%
  mutate(price = if_else(price_type == "Weekly", (price*52)/12, price)
  ) %>%
```

```
  mutate(price_type = if_else(price_type == "Weekly", "Monthly", price_type))
aptDataWorking <- filter(aptDataWorking, price_type != "Monthly|Weekly")
```

Discarding unwanted variables

We probably can't use:

- price_type (now the same for all individuals)
- USD (also same for all individuals)
- title: looks like mostly just basic info (bedrooms) and address, we already have that information stored elsewhere
- body: massive block of text which we won't have the tools to analyze until we take DATA 322
- address: again, we don't have tools to analyze text, and there isn't any real information contained here
- time: seems to describe hour of day ad was accessed, not useful
- price_display: just a less useful version of price (can use ggplot commands to display nicely)
- id (contains no information)

We also remove pets_allowed, as it wasn't important for any of our research questions and contains many NA values.

```
aptDataWorking <- select(aptDataWorking, -price_type, -title, -body, -time, -
price_display, -address, -currency, -id, -pets_allowed)
colnames(aptDataWorking)
```

```
 [1] "category"    "amenities"   "bathrooms"   "bedrooms"    "fee"
 [6] "has_photo"   "price"       "square_feet" "cityname"    "state"
[11] "latitude"    "longitude"   "source"
```

Renaming cityname

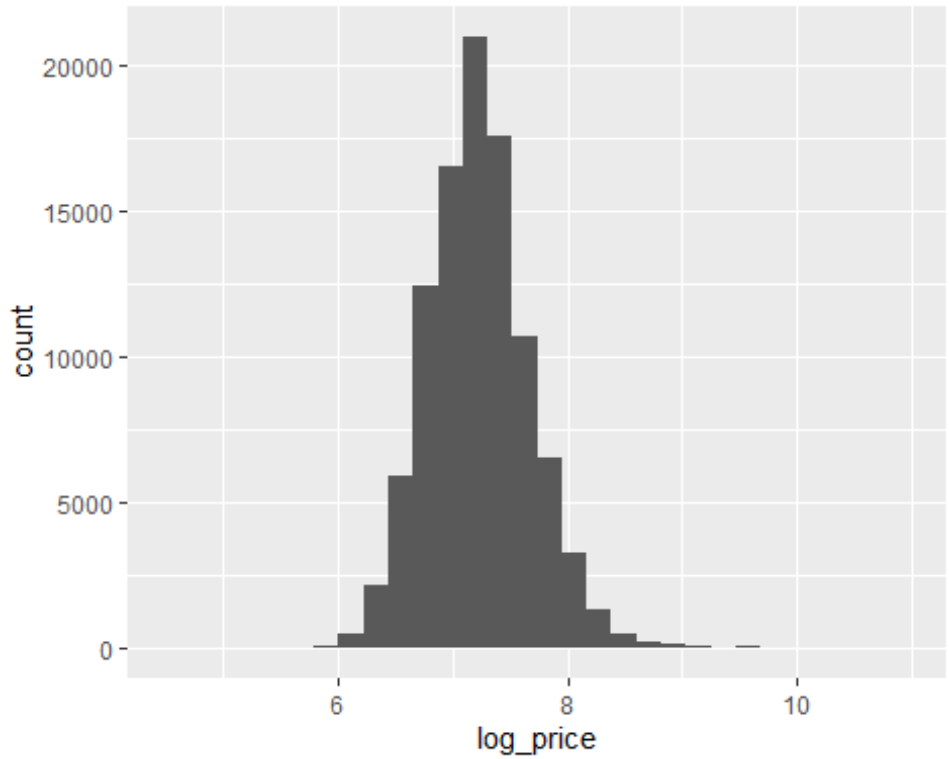
cityname is the only variable that doesn't match the underscore format.

```
aptDataWorking <- aptDataWorking %>% rename(city_name = cityname)
```

Logarithmically transforming price and square feet

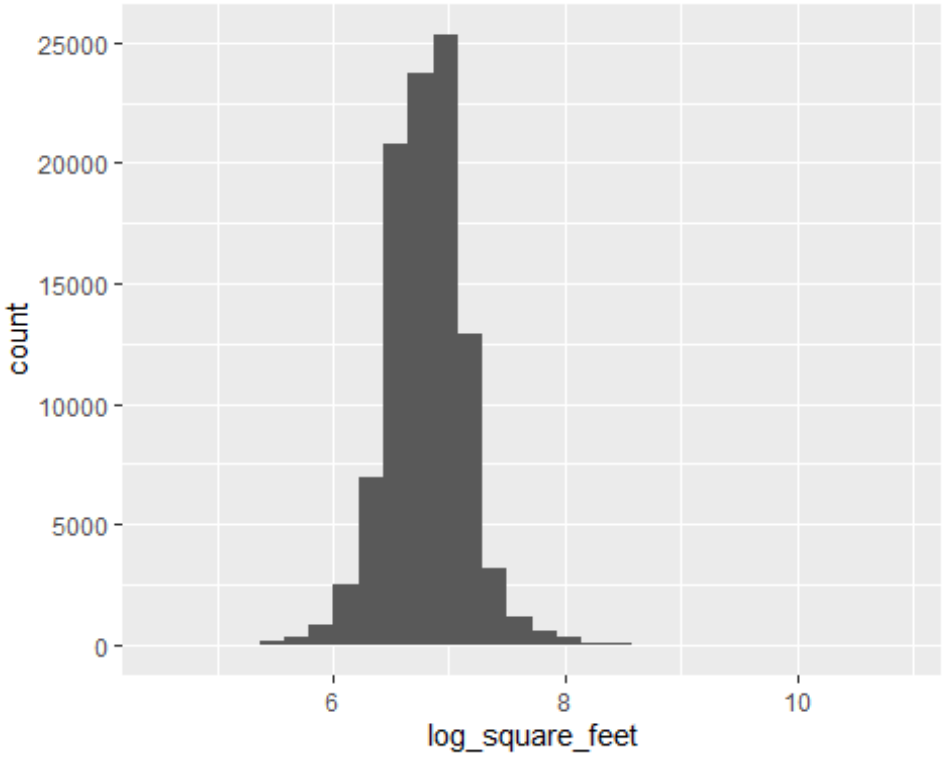
```
aptDataWorking <- aptDataWorking %>% mutate(log_price = log(price))
aptDataWorking %>%
  ggplot() +
  geom_histogram(mapping = aes(x=log_price))
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
aptDataWorking <- aptDataWorking %>% mutate(log_square_feet = log(square_fee
t))
aptDataWorking %>%
  ggplot() +
  geom_histogram(mapping = aes(x=log_square_feet))

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



Making categorical variables factors

```
aptDataWorking <- aptDataWorking %>%
  mutate(fee = as.factor(fee)) %>%
  mutate(has_photo = as.factor(has_photo)) %>%
  mutate(cityname = as.factor(city_name)) %>%
  mutate(state = as.factor(state)) %>%
  mutate(source = as.factor(source))
aptDataWorking
```

```
# A tibble: 98,923 × 16
   category       amenities bathrooms bedrooms fee   has_photo price square_f
eet
   <chr>          <chr>     <chr>        <dbl> <fct> <fct>     <dbl>       <d
bl>
 1 housing/rent/… null      1                1 No    Thumbnail  2195
542
 2 housing/rent/… null      1.5              3 No    Thumbnail  1250        1
500
 3 housing/rent/… null      2                3 No    Thumbnail  1395        1
650
 4 housing/rent/… null      1                2 No    Thumbnail  1600
820
 5 housing/rent/… null      1                1 No    Thumbnail   975
624
 6 housing/rent/… null      1.5              2 No    Thumbnail  1250
965
```

```
 7 housing/rent/… null      2                2 No    Thumbnail  1600        1
120
 8 housing/rent/… null      2                2 No    Thumbnail  1300
947
 9 housing/rent/… null      1                2 No    Thumbnail   795
600
10 housing/rent/… null      2                2 No    Thumbnail  2150        1
005
# ℹ 98,913 more rows
# ℹ 8 more variables: city_name <chr>, state <fct>, latitude <dbl>,
#   longitude <dbl>, source <fct>, log_price <dbl>, log_square_feet <dbl>,
#   cityname <fct>
```

Making bed/bath numeric

```
aptDataWorking <- aptDataWorking %>%
  mutate(bathrooms = as.numeric(bathrooms)) %>%
  mutate(bedrooms = as.numeric(bedrooms))
```

Separating amenities

Each value in amenities is a comma-separated list of different amenities (gym, dishwasher, pool, etc). The code below creates a column for each amenity, which contains the TRUE boolean if an observation has that amenity listed in amenities. To ensure all amenities were covered, the copy amenities1 of amenities was created, and separated into 20 columns, each containing one amenity (observations with less than 20 amenities have some columns filled with NA), which were all read with unique to obtain examples of all possible amenities values.

```
aptDataWorking <- aptDataWorking %>%
  mutate(amenities1 = amenities) %>%
  separate(amenities1, into=c("am1", "am2", "am3", "am4", "am5", "am6", "am7",
 "am8", "am9", "am10", "am11", "am12", "am13", "am14", "am15", "am16", "am17",
 "am18", "am19", "am20"), sep=",")
unique(aptDataWorking$am1)
```

```
 [1] "null"               "Fireplace"          "Gated"
 [4] "Dishwasher"         "Gym"                "Clubhouse"
 [7] "Refrigerator"       "Cable or Satellite" "Parking"
[10] "Basketball"         "Pool"               "Elevator"
[13] "Patio/Deck"         "Garbage Disposal"   "Storage"
[16] "Tennis"             "Alarm"              "Internet Access"
[19] "Hot Tub"            "Wood Floors"        "View"
[22] "Playground"         "AC"                 "Washer Dryer"
[25] "Luxury"             "TV"                 "Doorman"
[28] "Golf"
```

```
aptDataWorking <- aptDataWorking %>%
  mutate(fireplace = grepl("Fireplace", aptDataWorking$amenities, fixed=TRUE))
 %>%
```

```
  mutate(gated = grepl("Gated", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(dishwasher = grepl("Dishwasher", aptDataWorking$amenities, fixed=TRU
E)) %>%
  mutate(gym = grepl("Gym", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(clubhouse = grepl("Clubhouse", aptDataWorking$amenities, fixed=TRUE))
 %>%
  mutate(fridge = grepl("Refrigerator", aptDataWorking$amenities, fixed=TRUE))
 %>%
  mutate(cable_or_sat = grepl("Cable or Satellite", aptDataWorking$amenities,
 fixed=TRUE)) %>%
  mutate(parking = grepl("Parking", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(elevator = grepl("Elevator", aptDataWorking$amenities, fixed=TRUE))
 %>%
  mutate(basketball = grepl("Basketball", aptDataWorking$amenities, fixed=TRU
E)) %>%
  mutate(pool = grepl("Pool", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(patio_deck = grepl("Patio/Deck", aptDataWorking$amenities, fixed=TRU
E)) %>%
  mutate(garb_disposal = grepl("Garbage Disposal", aptDataWorking$amenities,
fixed=TRUE)) %>%
  mutate(storage = grepl("Storage", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(tennis = grepl("Tennis", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(alarm = grepl("Alarm", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(internet = grepl("Internet Access", aptDataWorking$amenities, fixed=
TRUE)) %>%
  mutate(hot_tub = grepl("Hot Tub", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(wood_floors = grepl("Wood Floors", aptDataWorking$amenities, fixed=T
RUE)) %>%
  mutate(view = grepl("View", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(playground = grepl("Playground", aptDataWorking$amenities, fixed=TRU
E)) %>%
  mutate(AC = grepl("AC", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(laundry = grepl("Washer Dryer", aptDataWorking$amenities, fixed=TRU
E)) %>%
  mutate(luxury = grepl("Luxury", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(TV = grepl("TV", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(doorman = grepl("Doorman", aptDataWorking$amenities, fixed=TRUE)) %>%
  mutate(golf = grepl("Golf", aptDataWorking$amenities, fixed=TRUE)) %>%
  select(-amenities, -(am1:am20))
```

Changing RENTCaf to RENTCafe

```
aptDataWorking <- aptDataWorking %>%
  mutate(source = if_else(source == "RENTCaf\xe9", "RENTCafe", source)
  )
```

What should be “RENTCafe” is instead “RENTCaf”, presumably because of an inability to display a character with an accent mark. This replaces the accented e in every instance of “RENTCafe” in source with a regular e.

Write changes into clean csv

```
write.csv(aptDataWorking, "apartments_clean.csv")
```

Simple linear regression model

```
fit.log_square_feet <- lm(log_price ~ log_square_feet, data=aptDataWorking)
summary(fit.log_square_feet)
```

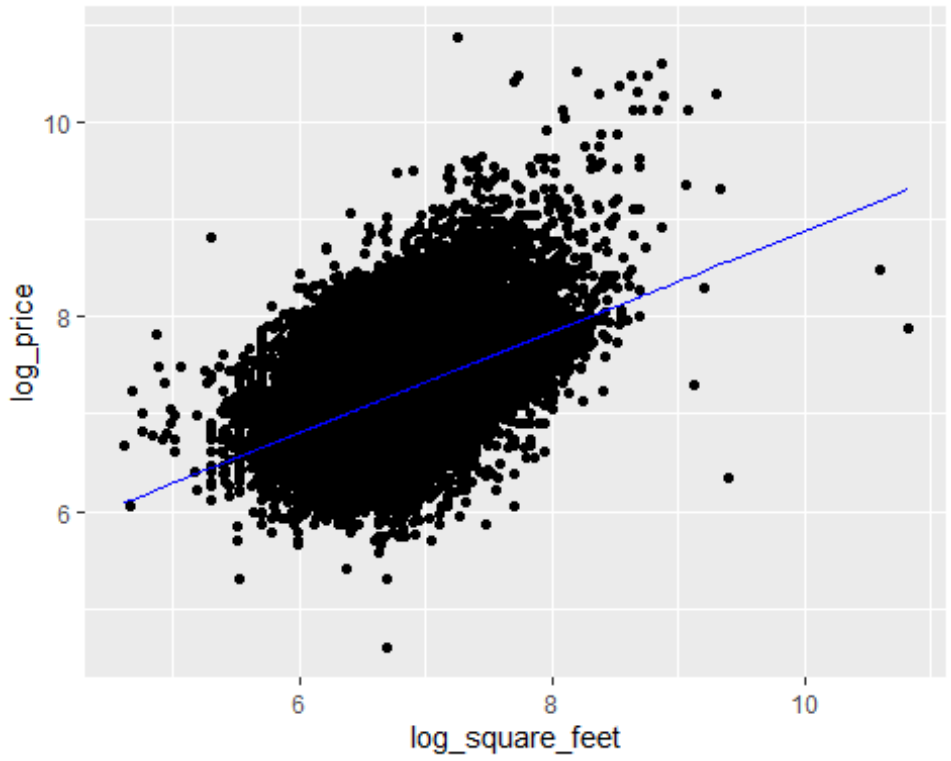
```
Call:
lm(formula = log_price ~ log_square_feet, data = aptDataWorking)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.5589 -0.2812 -0.0323  0.2490  3.4087 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)     3.709485   0.025975   142.8   <2e-16 ***
log_square_feet 0.516795   0.003813   135.5   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4037 on 98920 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.1566,    Adjusted R-squared:  0.1566 
F-statistic: 1.837e+04 on 1 and 98920 DF,  p-value: < 2.2e-16
```

```
aptDataWorking %>%
  add_predictions(fit.log_square_feet) %>%
  ggplot() +
    geom_point(mapping = aes(x=log_square_feet, y=log_price)) +
    geom_line(mapping = aes(x=log_square_feet, y=pred), color="blue")
```


10
8
6
log_price
6
8
10
log_square_feet